

Garment production and incentive system

Master plan, business rules, architecture and delivery plan

A tablet-first system, entirely in Amharic, that follows garments from raw fabric to delivery, pays a fixed salary plus a bi-weekly incentive for work over target, and sends management the same tables they read today, automatically, by Telegram and Google Drive.

Version 2.0 (revised), September 20, 2026. Based on the client's requirement notes, the two sample files (salary_schedule.pdf and incentive_v4.pdf) and the decisions made after version 1: a Super Manager and an Admin, fixed salaries, and incentive only for work over target. This document contains no code. It explains what is being built, how it behaves and why.

1	2	3	4	5	6	7	8	9
Receiving	Cutting	Sewing	Trimming	Quality control	Styling, hit press	Ironing	Packing	Delivery
ጥሬ እቃ መቀበያ	ቆረጣ	ስፌት	ለቀማ	ጥራት ቁጥጥር	ሂትፕረስ	ካውያ	ማሸግ	ማድረስ

Dashed, shaded stages (2 to 8) are the ones where workers can earn an incentive for work over target. Stages 1 and 9 are tracked but carry no incentive.

What changed in version 2

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2. What the two PDFs show

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4. Roles and permissions

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What changed in version 2

Version 2 applies the decisions made after version 1 was reviewed. Everything else from version 1 remains, except where these decisions made a rule obsolete.

Decision	What it means for the system	Where
One Super Manager and one Admin control everything	Two top-level roles replace the single Admin/Owner. The Super Manager manages all department roles and approves pay. The Admin controls users, settings and the system. There are now nine roles.	Chapter 4
Salary is fixed, not paid per piece	The wage piece rate and the daily piece-rate pay are removed. Every employee has a fixed salary. The daily salary schedule becomes a daily production sheet (no pay columns) plus a monthly fixed salary schedule.	2.1, 5.3, chapter 8
Incentive is paid only for work over target	The incentive card (target per hour and rate per piece) is the only rate table. The payable incentive is never below zero and nothing is deducted from the fixed salary.	5.4
Everything else remains	Nine stages, wastage alerts, penalty ladder, Telegram and Drive automation, Amharic interface, architecture and roadmap stay, apart from wording that followed from the three decisions above.	Chapters 3, 5 to 10
New in this version	A recommendations chapter with 22 items. The open questions are rewritten: two old questions are settled by the decisions above, and new ones cover the fixed salary, shortfalls and the Super Manager and Admin split.	Chapters 12, 13

Two interpretations to confirm. First, I read "incentive when work is over target" as: a shortfall never reduces pay, and the payable incentive is never below zero ([Q2](#)). Second, I gave the Super Manager the approval of pay and the Admin the control of the system, with both able to see and act on everything ([Q11](#)).

1. What you are building

The factory turns fabric into T-shirts and trousers in nine stages. The numbers behind that work (who sewed how many pieces, what incentive each person earned, how much fabric was wasted, what is in stock) are recorded on forms like the two sample PDFs. The system replaces those forms with tablet entry, does the arithmetic itself, and sends finished tables to management.

The three jobs of the system

1. **Record** what happens at every stage, by the person who does it, on a tablet or phone, in Amharic.
2. **Calculate** what the records mean: the incentive for work over target, the penalty rules, and fabric wastage.
3. **Deliver** alerts, the Amharic daily summary and the pay tables to management on Telegram, and archive every file in Google Drive, without anyone building a report by hand.

How people are paid

Every employee has a **fixed salary** that does not depend on how many pieces they make. On top of it, an employee can earn an **incentive** for pieces above the daily target, calculated over 15-day periods. This one fact drives much of the design: production counts matter because they decide the incentive, not the salary.

Goals

- One system for stock, cutting, production, quality, finishing and delivery.
- Incentive outputs that reproduce incentive_v4.pdf to the cent. The samples double as acceptance tests (chapter 10).

- Every screen, message and report in Amharic, usable on shop-floor tablets and smartphones.
- Full control for one Super Manager and one Admin, with a clear trail of who changed what.
- Simple enough that one developer can build, run and hand it over.

Scope

In scope

- Stages 1 to 9 tracking, by bundle
- Nine user roles, including a Super Manager and an Admin
- Employee list with fixed salaries, attendance and line assignment
- Hourly piece counts per worker
- Bi-weekly incentive statements, penalty ladder, monthly summary and chart
- Daily production sheet and monthly fixed salary schedule
- Fabric wastage calculation and alerts
- Stock levels and low-stock alerts
- Four core reports plus the Amharic daily summary
- Telegram delivery and Google Drive archive
- Barcode or tag for finished goods
- Audit log and backups

Out of scope

- Live truck tracking. Delivery records what left, not where it is.
- Accounting and invoicing
- Tax, pension, overtime and absence deductions, unless the client asks ([Q1](#))
- Purchase orders and supplier payments
- Customer portal

Items in this list are not in the requirement notes. They can become later phases, but they are not planned here.

How big is it?

The samples describe about 56 to 67 workers and 9 system users. If every worker gets an hourly count for 8 hours, that is roughly 540 count cells per day. This is a small data problem: one PostgreSQL database on modest hardware is more than enough. The hard part is not volume. It is **getting the incentive rules exactly right** and making entry fast enough that supervisors prefer it to paper.

How to read this document. Chapters 2 and 5 are the core: what the PDFs say and the rules the system must obey. Chapters 6 and 7 are the technical design. Chapter 12 gives recommendations and chapter 13 lists the questions to settle with the client before the pay engine is built.

2. What the two PDFs show

These two files are the specification for the most important outputs. Everything in this chapter was read out of them. I recomputed every row: each salary line equals produced pieces $\times$ piece rate, and each incentive line equals $(+ - - - 2 \times \text{mistakes}) \times \text{rate}$. Nothing in the samples contradicts the formulas. Because version 2 makes the salary fixed, the salary file is now mainly a source of layout, operations, machines and targets, and the incentive file is the source of the pay rules.

2.1 salary_schedule.pdf: the daily register, now without pay

Title: የሰራተኞች ደሞዝ ሰሌዳ (Employee Salary Schedule), a daily register that paid workers by piece rate. With a fixed salary, the pay columns go away, but the sheet remains valuable as the daily production record: one sheet per day, 56 workers in 13 operations, over two printed pages.

Keep	Drop	Add
Header: date (Ethiopian calendar, sample 2018/12/15), shift, supervisor, manager. Columns: serial number, name, operation, machine type, daily target, produced, signature. Rules: daily target = hourly target × 8 hours; rework pieces are not counted.	The piece rate column, the daily pay column, and the formula "daily pay = produced × piece rate".	Difference against target (+ or -) and percent of target for each worker, because the incentive is built on these.

Daily difference = pieces produced – (hourly target × hours worked)

A full shift is 8 hours. Helpers have no target on this sheet.

Structure of the sample day, with figures I calculated from the rows:

#	Operation	Machine	Workers	Target per day (per hour)	Produced	At or above target
1	Cutting and preparation, helpers (ቆረጣ እና ዝግጅት)	Cutter or helper	5	520 (65)	2,622	3 of 5
2	DTF sticker heat-pressing	Hitpress	4	656 (82)	2,656	3 of 4
3	Pocket sewing and topstitch (ኪስ መስፋት + ደርዝ)	Single needle	3	440 (55)	1,348	3 of 3
4	Neck rib and back tape	Single needle	2	656 (82)	1,331	1 of 2
5	Waistband and elastic joining	Single or overlock	3	440 (55)	1,331	3 of 3
6	Waist, Kanshay (ወብ በካንሻይ መምታት)	Kanshay	1	1,304 (163)	1,299	0 of 1
7	Trouser panel seam, front and back	Overlock	2	656 (82)	1,337	2 of 2
8	Shoulder and sleeve joining	Overlock	6	440 (55)	2,672	4 of 6
9	T-shirt and trouser side seam	Overlock	8	328 (41)	2,674	6 of 8
10	T-shirt and trouser hem or sleeve fold	Interlock	4	328 (41)	1,342	3 of 4
11	Line material feeder or helper	Helper	3	none	none	none
12	Thread trimming (ክር ለቃሚ)	Trimmer	11	240 (30)	2,718	9 of 11
13	Ironing and packing (ካውያ እና ማሸግ)	Kawya	4	328 (41)	1,349	3 of 4
Total			56		22,679	40 of 53

Two follow-ups. The fixed salary schedule has no sample yet, so its layout is unknown: ask the client for one (Q1). The old note about different T-shirt and trouser piece rates on operations 8, 9 and 10 was about wages; whether incentive rates also differ by garment is Q13.

2.2 incentive_v4.pdf: the bi-weekly incentive statement

This is a multi-part document. Each part becomes a screen, a report, or both. Its own rules page says the incentive is paid separately from the monthly salary, which matches the fixed-salary decision.

- 1. **Rules page.** The special note that incentive is paid separately from the monthly salary; the two pay dates (day 4 and day 19); the formula; motivational lines for workers; and the four-step penalty ladder.

- Rate card.** 21 rows: department, machine type, hourly target and incentive rate per piece (ሳንቲም, in ETB). Now the only rate table. Reproduced in the appendix.
- Statement for day 4.** 67 workers. Columns: serial, name, department, machine, rate, +pieces, –pieces, mistakes (×2), incentive (ETB), signature. Header: አንስታሊክ ክፍያ - ቀን 4, counting period from day 20 of the previous month to day 4. Total 3,880.50 ETB.
- Statement for day 19.** Same layout. Counting period day 5 to day 19. Total 3,765.00 ETB.
- Daily piece-count form.** The blank sheet supervisors fill: date, supervisor, department, target per hour, eight hourly columns, daily +/- . It feeds the incentive statement. This is what the tablet screen replaces.
- Monthly summary and chart.** 24 departments with day-4 total, day-19 total and month total, plus a bar chart of monthly incentive by department. Grand total 7,645.50 ETB.

$$\text{Calculated incentive (ETB)} = (+\text{pieces} - -\text{pieces} - 2 \times \text{mistakes}) \times \text{rate per piece}$$

In the sample, negative results are printed as negative numbers. Version 2 adds a Payable amount that is never below zero (section 5.4).

Worked lines from the sample

Worker and department	Rate	+	–	Mistakes	Calculation	Calculated	Payable
Senait Teferi, Overlock, shoulder (period 1)	1.00	200	0	0	$(200 - 0 - 0) \times 1.00$	200.00	200.00
Sewnet Kabhun, Interlock hem (period 1)	1.20	100	0	0	$(100 - 0 - 0) \times 1.20$	120.00	120.00
Shewaye Shimeles, pocket attaching (period 1)	0.90	0	30	0	$(0 - 30 - 0) \times 0.90$	–27.00	0.00
Constructed example, because the sample column is always 0	0.90	200	20	5	$(200 - 20 - 10) \times 0.90$	153.00	153.00

Penalty ladder printed on the rules page

- First offence:** twice the mistaken count is deducted from the incentive.
- Second offence:** incentive is fully suspended for two months.
- Third offence:** written warning and disciplinary action on the job.
- Fourth offence:** action up to termination under company regulations.

My reading of "ጭሳሳት" (mistake). The rules page says accuracy of numbers is the basis of trust, so I read a "mistake" as a piece count that was reported wrongly (false or miscounted), not a sewing defect. This is a guess to confirm. See [Q6](#).

2.3 What the two PDFs still disagree on

The two files do not describe the same list of people or the same targets. The system must handle both without forcing them together.

Topic	salary_schedule.pdf	incentive_v4.pdf	Consequence
People	56 names	67 names, different names	One employee master. Each person belongs to one department. The samples are probably from different files, so nothing can be merged by name.
Grouping	13 operations	21 rate rows, 24 summary departments	Two classification schemes. A mapping between them is needed (Q3).

Topic	salary_schedule.pdf	incentive_v4.pdf	Consequence
Target	Hit press 656 per day (82 per hour), ironing and packing 328 (41)	Hit press 90 per hour, ironing 45, packing 55	Targets differ for the same work. Version 2 uses the incentive card as the single source (Q3).
Rhythm	Daily	Every 15 days, days 4 and 19	A daily production sheet plus period statements.
Ironing and packing	One combined operation	Two departments	Departments must be finer-grained than operations.

A few decorative banner lines on the incentive rules page did not extract legibly from the file, so I did not rely on them. If they contain rules, send the original wording.

3. How the factory flows

A cut bundle is the unit that travels. Workers' counts are recorded separately per person per hour, because the incentive is calculated from them. Both views matter: bundles show where work is, counts show who did it and what they earn.

#	Stage	Who records	What is recorded	Result	Earns incentive?
1	Raw material receiving	Store Keeper	Scan or enter arrival: material, supplier, lot, quantity, unit. The system assigns the SKU.	Stock updated. Low-stock check.	No
2	Issue and cutting	Cutting Manager (requests), Store Keeper (issues)	Fabric requested per order. After cutting: total weight used, pieces cut, bundles created.	Wastage % calculated at once. Alert above 5%. Bundles released to sewing.	Yes
3	Sewing	Operators and Production Manager	Bundles arrive at lines. Hourly piece counts per worker.	Live progress for management.	Yes
4	Trimming (ለቀማ)	Supervisor	Pieces trimmed per worker.	Bundle moves on.	Yes
5	Quality control	QC Inspector	Inspect, pass or fail, defect reason, responsible operation.	Passed items go to the finished-goods staging queue. Defective items are scanned and sent back to the specific line for repair.	Yes
6	Styling, hit press	Supervisor	Pieces pressed per worker.	Bundle moves on.	Yes
7	Ironing (ካውያ)	Supervisor	Pieces ironed per worker.	Bundle moves on.	Yes
8	Packing (ፓኪንግ በማዳበሪያ)	Supervisor or Finished Goods Manager	Pieces packed into bags per worker.	Finished goods ready.	Yes
9	Delivery	Finished Goods Manager	Quantity dispatched, customer, date, tag or barcode.	Order status updated. No live truck tracking.	No

Order of stages. As listed, quality control (5) comes before styling, ironing and packing (6 to 8). Many factories inspect again at the end. The system should not hard-code the order: each style gets an ordered **route** of stages, defaulting to the nine above, so a style without hit press, or with a final check, needs no redesign. See [Q10](#).

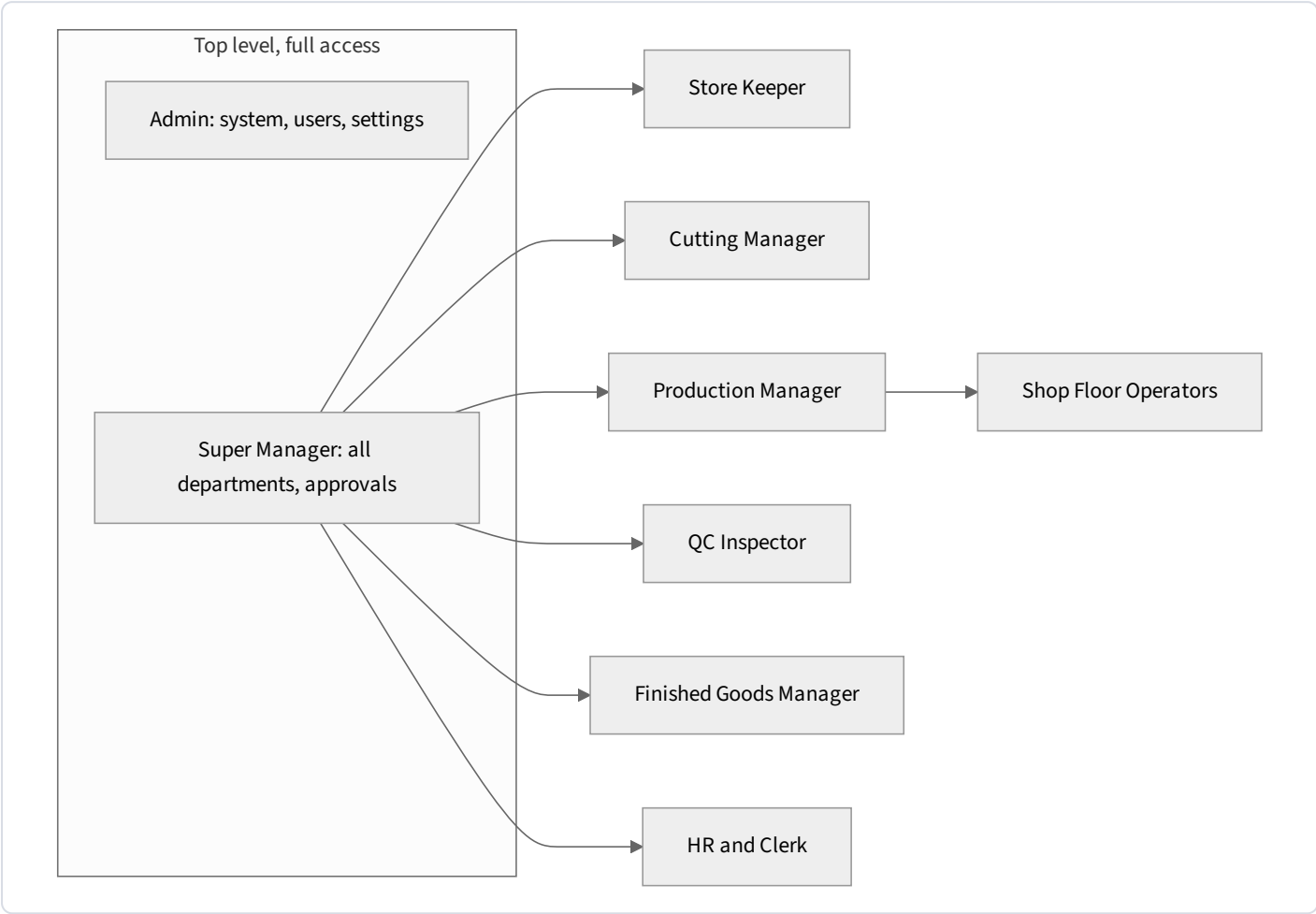
Which incentive departments belong to which stage

The incentive rate card is finer than the nine stages. This is my reading of how its departments map to stages (English names are approximate; the Amharic terms are authoritative).

Stage	Departments on the rate card	Workers in sample
2 Cutting	ቆራጭ cutter (1), ረዳት ቆራጭ assistant cutter (1), ስቴከር / ላስቲክ መቀጥ sticker and elastic cutting (1), ረዳት / ሱሪ መቁረጥ helper, trouser cutting (1)	4
3 Sewing	ትከሻ (በክ) ፍሬንት shoulder back and front (4), ወገብ መቀምቀም waist tacking (1), እጅ / ሳይድ sleeve and side (4), ኪስ መለጠፍ pocket attaching (4), ሳይድ side (4), እጅት sleeve (4), እጅት ደርዝ እና ባጅ sleeve topstitch and badge (2), Interlock ማጠፍ interlock hem (4), ወገብ መቀጠም waistband joining (4), የካንሻይ ወገብ ዝግጅት Kanshay waist prep (3)	34
4 Trimming	ቅንጫባ (ክር ለቃሚ) thread trimming (8)	8
5 Quality control	Quality inspector (1), ዳሜጅ / ሱሪ መስረት damage and repair (2)	3
6 Styling	ሂትፕረስ መለጠፍ heat-press applying (4)	4
7 Ironing	ካውያ ironing (4)	4
8 Packing	ማሸግ packing (5)	5
Support	የመስመር ረዳት line helpers (2), ረዳት helper (1), መስመር ማሰራት line running (1), ማሰራት / ቁጥጥር line control (1). Target 0, so their incentive is always 0.	5
Total		67

4. Roles and permissions

The client asked for at least eight roles and, in version 2, for two people at the top: one **Super Manager** who manages all the other roles, and one **Admin**. Both control everything. The other seven roles come from the requirement notes. The permission table is configuration, not code, so it can be adjusted later.



Role	Amharic	Main duty
Admin	አስተዳዳሪ	Controls the system: creates users, assigns roles and permissions, sets alert rules and connections (Telegram, Google Drive), runs backups and reads the audit log. Can correct locked data, with a required reason.
Super Manager	ሱፐር ማኔጀር (ዋና ሥራ አስኪያጅ)	Manages all the other roles and departments. Sees every screen and report, approves incentive statements, salary schedules and salary changes, sets targets and incentive rates, and receives all Telegram reports and alerts.
Store Keeper	የመጋዘን ኃላፊ	Records material in and out and reports stock levels.
Cutting Manager	የቆረጣ ኃላፊ	Requests fabric, records weight used, pieces cut, bundles and wastage.
Production Manager	የምርት ኃላፊ	Follows sewing and the daily target, verifies counts. Likely the same person as the "supervisor" on the forms (Q12).
QC Inspector	የጥራት ተቆጣጣሪ	Checks quality, identifies defects, sends repairs back.
Finished Goods Manager	የተጠናቀቀ እቃ ኃላፊ	Receives finished garments, tags them, dispatches to customers.
HR / Clerk	የሰው ሀብት / ጸሐፊ	Employee records, attendance hours, line assignment.
Shop Floor Operator	የሰራ ቦታ አገልግሪ	Reports daily work quantity by tablet or smartphone.

The two top roles side by side

Both roles can open every screen and act in any department, and every such action is marked "entered by manager" in the audit log. What differs is who holds which key. These are my defaults, to confirm in [Q11](#).

Topic	Admin	Super Manager
Purpose	Keep the system running, secure and correctly configured.	Run the business through the department roles.
Users, roles, permissions	Creates and changes them.	Views them.
Targets, incentive rates, fixed salaries	Can edit, but each change notifies the Super Manager.	Sets and approves them.
Approve incentive statements and salary schedules	Views.	Approves. Nobody else can.
Correct locked data	Yes, with reason. Pay-related corrections notify the Super Manager.	Yes, with reason.
Connections, backups, settings	Controls.	Views the audit log and reports.
Telegram	Receives summary, alerts, pay files and system errors.	Receives summary, alerts and pay files.

Permission matrix

V = views, E = views and edits, A = views, edits and approves or corrects after lock. Blank = no access. Checks happen on the server for every request, not only by hiding buttons.

Function	Admin	Super Mgr	Store	Cutting	Prod. Mgr	QC	Fin. Goods	HR	Operator
Users, roles, permissions, settings, connections, backups	A	V							
Employee master (without salary)	E	A			V			E	
Fixed salary amounts	E	A							
Incentive card: targets and rates	E	A			V				
Material receiving, stock issue	E	E	E	E*	V				
Cut records and wastage	E	E	V	E	V				

Function	Admin	Super Mgr	Store	Cutting	Prod. Mgr	QC	Fin. Goods	HR	Operator
Bundle progress through stages	E	E		V	A	V	V		E
Hourly worker counts	E	A			A			V	E**
Attendance and line assignment	E	E			V			E	V
QC inspections and defects	E	E			V	E	V		
Finished goods and delivery	E	E			V	V	E		
Incentive adjustments, mistakes, offences	E	A		E***	E***	E***			
Approve incentive statements and salary schedules	V	A							
Dashboards and reports	V	V	V	V	V	V	V	V	
Audit log	V	V							

- * Cutting Manager creates fabric requests; the Store Keeper fulfils them.
- ** Operator entries stay "pending" until the supervisor verifies them. The person whose incentive depends on a count should not be its only source.
- *** Only for their own departments (for example, Cutting Manager for cutters). The client's notes say the admin or the department role enters + and -. The Super Manager approves before the statement is final.

Two keys on money. Whoever enters a count does not verify it, and whoever verifies it does not approve the statement: supervisor enters, Production Manager verifies, Super Manager approves. Fixed salaries are visible only to the Super Manager and the Admin. HR sees attendance but not salary unless the client decides otherwise ([Q15](#)).

Every user sees only their own role's screens, in Amharic. Shared tablets use a short login (employee ID plus PIN) with automatic sign-out after inactivity.

5. Business rules

This chapter is the contract for the calculation module. Each rule includes a worked example so it can become an automated test. Where the PDFs do not settle a rule, a flag points to the open question and the default I would build.

5.1 Cutting wastage

Wastage % = (weight used – pieces cut × standard weight per piece) ÷ weight used × 100

Standard weight per piece comes from the style's bill of materials. Calculated the moment the Cutting Manager saves the cut record.

Example: 100.0 kg used, standard 0.250 kg per piece, 376 pieces cut. Standard weight = 94.0 kg, waste = 6.0 kg, wastage = **6.0%**. That is above 5%, so a Telegram warning is sent at once. The 5% limit is a setting, per style if needed.

Confirm the method ([Q9](#)). The notes say the manager enters weight used and pieces cut. That only yields a wastage figure if a standard weight per piece exists. If the client instead weighs leftover offcuts, the formula changes to offcut weight ÷ weight used.

5.2 Targets, hours and hourly counts

- Each worker has an hourly target from the incentive card of their department.
- Daily target = hourly target × hours worked. A full shift is 8 hours, as the salary schedule states. Attendance provides the hours (example: 5 hours worked at 82 per hour gives a target of 410).
- Supervisors enter the number of pieces per hour (columns hour 1 to hour 8 on the paper form). The system derives the difference against target per hour and per day.
- Daily difference is positive or negative. It feeds the period's **+ pieces** or **– pieces**.

The paper form does not show whether the hourly cells hold actual pieces or differences. Entering actual pieces is safer and removes hand arithmetic. Confirm in [Q4](#), and confirm the absence rule in [Q5](#).

5.3 Fixed salary

Salary = a fixed amount per employee

It does not change with the pieces produced. Production only affects the incentive.

- Each employee has one fixed salary amount with an effective date. Changes keep history, so an old month always shows what was true then.
- Only the Super Manager and the Admin can see or change salaries. The Super Manager approves changes; any Admin change sends a notice to the Super Manager.
- The monthly fixed salary schedule lists name, department, fixed salary, days attended (from attendance, for reference) and a signature column. Its final layout needs a sample from the client.
- Not calculated unless the client asks: tax, pension, overtime and deductions for absence.

Version 1 left base payroll out of scope. Printing a fixed amount is easy; any calculation beyond that adds rules and legal duties. Confirm in [Q1](#).

5.4 Incentive for work over target

Calculated = (total +pieces – total –pieces – 2 × mistakes) × incentive rate

Payable = the larger of 0 and Calculated, and 0 during a suspension

Only work over target earns money. A shortfall never reduces the fixed salary.

- Accumulated over a 15-day counting period and paid on day 4 and day 19.
- Only first-time-good pieces count. Rework, and pieces QC has rejected, are excluded until repaired ([Q14](#)).
- Each statement prints two columns per worker: **Calculated** exactly as in the sample, negatives included, and **Payable**. The month summary and its chart use Payable.
- Each line is rounded to two decimals, then the period total is the sum of the printed lines. In the sample, totals equal the sum of the lines exactly.
- Money is stored as exact decimals, never as floating-point numbers, so cents never drift. The rate used on each line is copied onto it when the period closes, so later rate changes never alter old statements.

Sample totals	Period 1 (day 4)	Period 2 (day 19)	Month
Pieces above target (+) and below (-)	+4,865 and -430	+4,610 and -270	
Calculated (as printed in the sample), ETB	3,880.50	3,765.00	7,645.50
Payable (negatives as 0), ETB	4,243.50	4,009.00	8,252.50

Flooring at zero raises the sample month by 607.00 ETB, about 7.9%. The Super Manager should see this cost before approving the rule (recommendation R3).

Netting or surplus only (Q2). In the sample, no worker has both + and - in the same period, so netting and surplus-only give the same totals there. They differ when a worker has a good day and a bad day in the same period. Example at rate 0.90: +200 on one day and -100 on another. Netting pays $(200 - 100) \times 0.90 = 90.00$. Surplus-only pays $200 \times 0.90 = 180.00$. Default: net per period, as the sheet's formula does, then floor at zero.

5.5 Penalty ladder in the system

Offence	Rule from the sheet	What the system does
1st	2× the mistaken count deducted from incentive	Automatic through the mistakes term in the formula. It reduces the incentive only, never the fixed salary. The offence is recorded with reason, date and who entered it.
2nd	Incentive suspended for 2 months	Creates a suspension with start and end dates (two months, about four pay periods). Earned amounts still print but are marked suspended and payable 0.
3rd	Written warning and action on the job	Opens an HR case and notifies the Super Manager and Admin. No automatic pay effect.
4th	Action up to termination by regulation	Opens an HR case and notifies the Super Manager and Admin. No automatic pay effect.

The system records and recommends. People decide anything beyond the incentive amount. See [Q7](#) for how offences are counted over time and [Q6](#) for what a mistake is.

5.6 Pay calendar

- Ethiopian months have 30 days, except Pagume, which has 5 or 6. Period 1 runs from day 20 of the previous month to day 4 ($11 + 4 = 15$ days). Period 2 runs from day 5 to day 19.
- Edge case: for Meskerem the previous month is Pagume, which has no day 20. Period boundaries therefore live in configuration per month, not in a fixed formula ([Q8](#)).
- The system stores dates in one standard form and converts to the Ethiopian calendar for screens, reports and period logic. Time zone is East Africa Time (UTC+3, no daylight saving).

5.7 Alerts

Alert	Trigger	Goes to	Repeat rule
Low stock	On-hand quantity at or below the material's minimum level	Super Manager, Admin, Store Keeper	Once per day until restocked
Wastage	Cut job wastage above the limit (default 5%)	Super Manager, Admin, Cutting Manager, Production Manager	Immediately, once per cut job
Delayed order	Order behind its planned stage or past its due date	Super Manager, Admin, Production Manager	Daily until resolved
Report delivery failed	Telegram or Drive upload fails after retries	Developer chat, Admin	Each failure

Alert	Trigger	Goes to	Repeat rule
Missing day close (suggested)	Day not closed by an agreed time	Supervisor, Super Manager	Once
Unusual count (suggested)	A worker's count far above target (for example, above 150%)	Supervisor	On entry, needs confirmation

The last two are additions that protect incentive accuracy, since the incentive depends on counts.

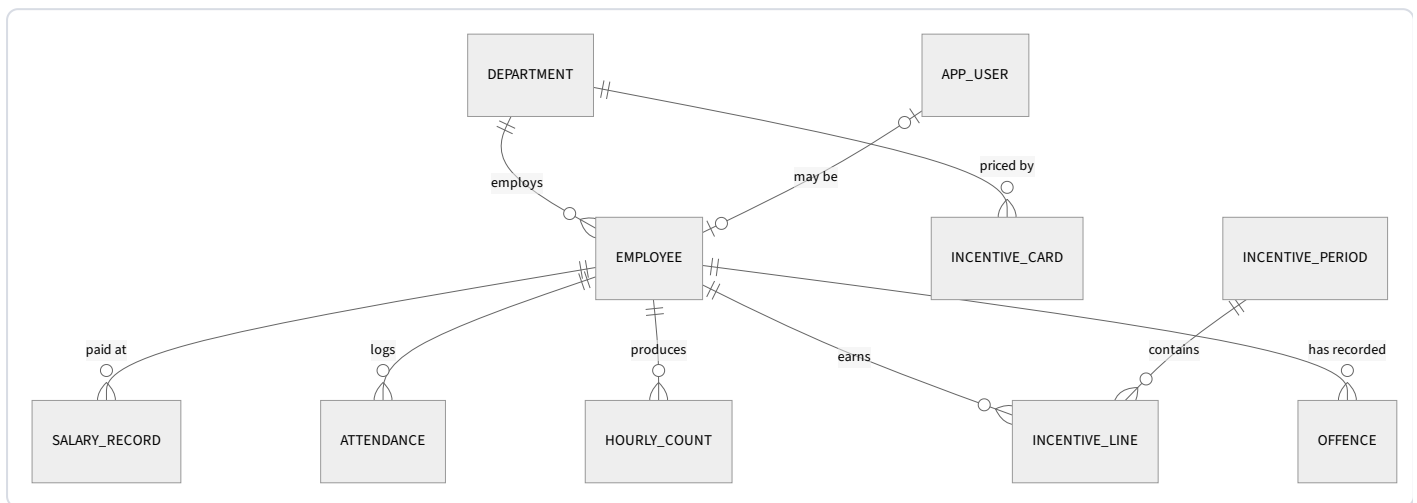
5.8 Locking and corrections

- Count entries move through **draft, submitted, verified, locked**. Closing the day locks it.
- After a lock, only the Admin or the Super Manager can change data, with a required reason. The old value stays in the audit log with who, when, before and after. Corrections by the Admin to pay data also notify the Super Manager.
- If a correction changes a report already sent, a new version is generated and the Telegram message says it replaces the earlier one. File names carry a version suffix.
- An incentive period cannot be locked until the Super Manager approves the statement.

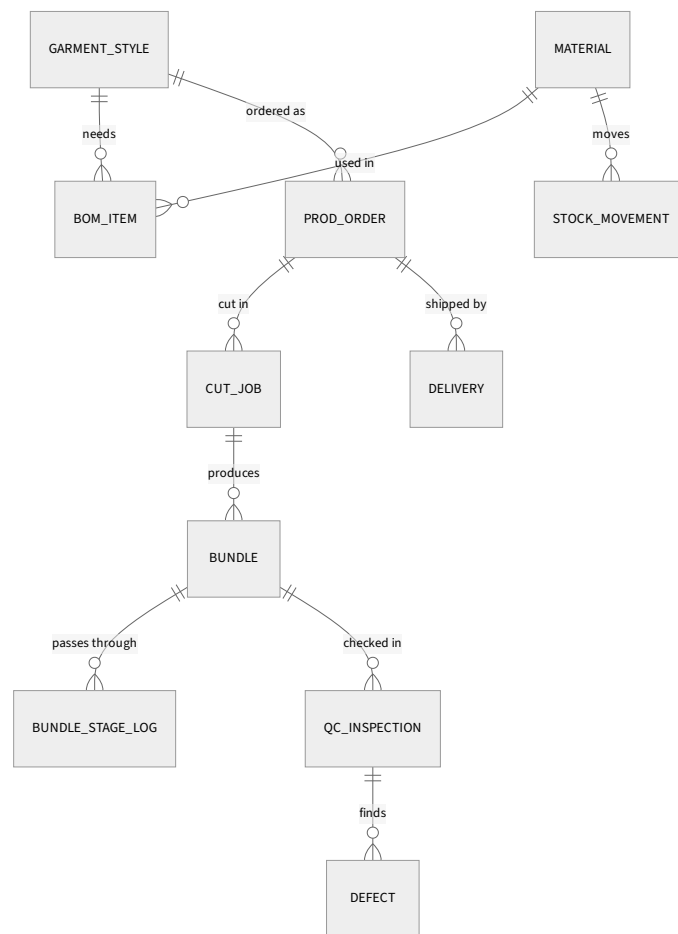
6. Data model

The data model is described in words and two relationship diagrams. The exact table definitions are a build-time task.

People and pay



Product and production



Domain	Main records	Notes
People	User (login and role, including Super Manager and Admin), Employee (ID, name, department, status), Department, Attendance (date, in and out, hours, line, shift), Line assignment	An employee may or may not have a login. Operators log in with ID and PIN.
Salary	Salary record: employee, fixed amount, effective from and to	Field-level access: Super Manager and Admin only. History kept.
Incentive card	Department or operation, target per hour, incentive rate per piece, effective from and to	The only rate table. Effective-dated so last month's statement still uses last month's rates.
Materials	Material or SKU (type, unit, minimum level), Supplier, Lot, Stock movement (receive, issue, return, adjust)	Stock on hand is the sum of movements, so history is never overwritten.
Product	Style (name, size, color), Bill of materials, Order (customer, style, quantity, due date, route of stages)	Master data from the requirement notes: SKU, style, size, color, unit, unit price, lot number, supplier, BOM.
Production	Cut job (weight issued, weight used, pieces, wastage %), Bundle (ID, lot, size, quantity), Bundle stage log, Hourly count, Day close	Bundle logs give independent totals to check worker counts against.
Quality	QC inspection, Defect (type, responsible operation and worker, repair status)	Rework is tracked so it can be excluded from incentive counts.
Delivery	Finished item or tag, Delivery (order, quantity, date, customer)	No vehicle tracking.
Incentive	Incentive period, Incentive line (plus, minus, mistakes, rate copy, calculated amount, payable amount, status), Offence, Suspension	Statement versions are kept when corrected. Approval by the Super Manager is recorded.
Automation	Alert, Report file (type, period, Drive file, Telegram message, status, version), Recipient (Telegram chat and reports allowed), Setting	Report file rows also power a "delivery status" screen.

Domain	Main records	Notes
Control	Audit log (who, when, what, before, after, reason, acted as manager or not)	Append-only.

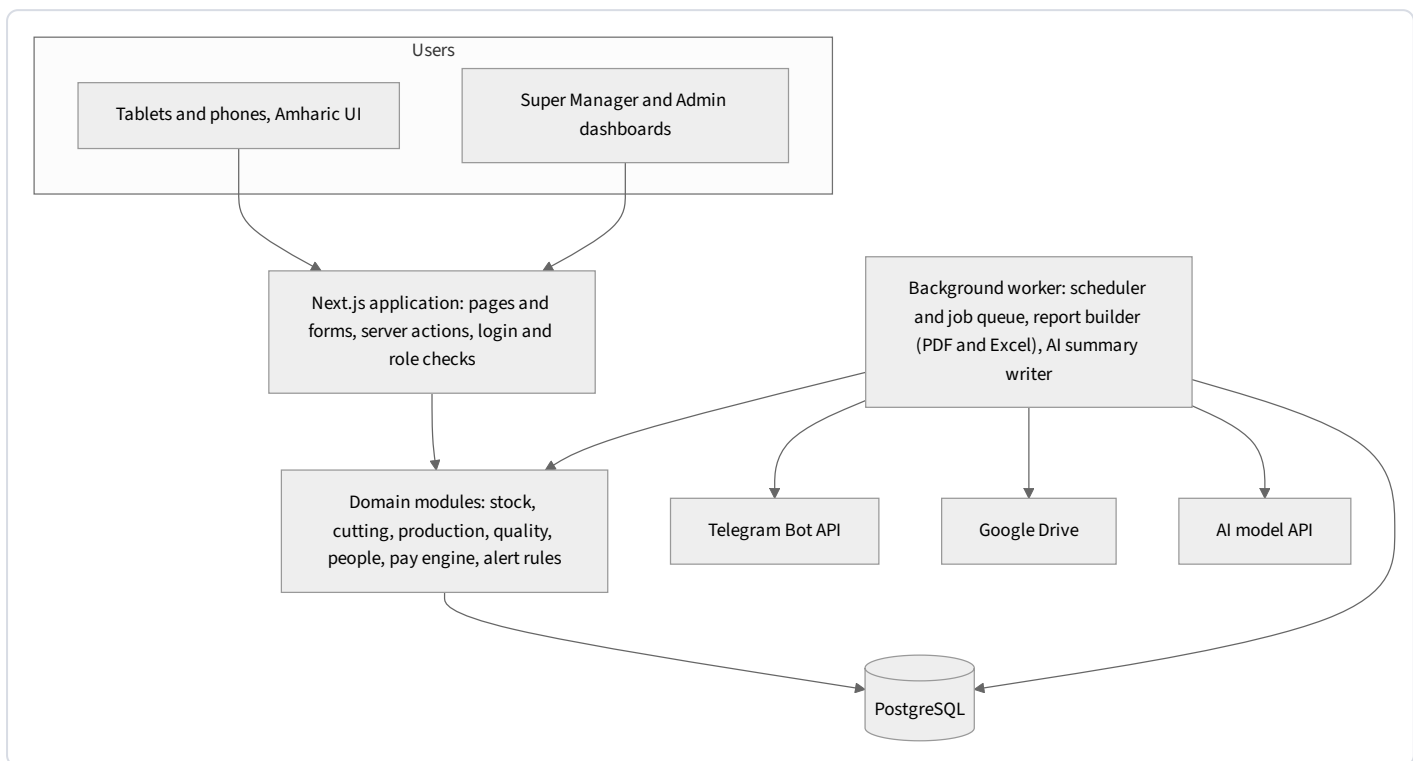
Design rules

- Never hard-delete business records. Mark them inactive.
- Every changeable value that affects money is dated and traceable.
- Client-generated IDs on entries, so a retried submission from a tablet never creates a duplicate.
- Reconciliation check: total of worker counts for a department vs the pieces logged for bundles at that stage. Large gaps are flagged, which also supports the "mistake" rule.

7. System architecture

7.1 Approach

A **modular monolith**: one Next.js application containing the user interface and the server logic, one PostgreSQL database, and one small background worker for scheduled and heavy jobs. Modules are separated inside the codebase (stock, cutting, production, quality, people, pay engine, alerts, reports) but deployed as a single unit. For a team of one, this is easier to build, test, host and hand over than separate services, and the data volume does not need more.



7.2 Technology choices

Layer	Choice	Why	Watch out for
App and API	Next.js with TypeScript	One framework for screens and server logic, as the client's stack notes specify.	Keep business rules in plain modules, not inside page components, so they can be tested alone.
UI	Tailwind CSS and shadcn/ui, Recharts for graphs, installable web app (PWA)	Responsive on tablets and phones. Recharts draws the department bar chart.	Large touch targets, number keypad inputs, high contrast for workshop light.

Layer	Choice	Why	Watch out for
Database	PostgreSQL with Prisma	Reliable relational storage and safe migrations.	Use exact decimal types for money and quantities.
Login	Auth.js with username and PIN, or a hosted provider such as Clerk	Workers have no email, so credentials login fits. Hosted auth is quicker to start but adds a vendor and possible cost.	PIN rules, lockout after failed attempts, sign-out on shared tablets.
Background jobs	A worker process with a Postgres-backed queue, or platform cron plus a report outbox table	Daily close, period close, retries.	Serverless platforms limit run time and cron frequency.
PDF and Excel	HTML-to-PDF with a headless browser, or a PDF library; a spreadsheet library for Excel	The sample layouts are wide tables with Amharic text.	Embed an Ethiopic font in every PDF. A headless browser gives the best layout fidelity but needs a host that can run it.
Notifications	Telegram Bot API	Free, works on phones, supports files.	Bot messages are not end-to-end encrypted. Message length limit about 4,000 characters. File size limit about 50 MB.
Archive	Google Drive API with narrow, per-file access	Client already uses Drive. Owner can browse the archive.	Service-account files are owned by that account; owner-authorized access counts against the owner's storage. Choose deliberately.
AI	A hosted language model (a free tier such as Gemini is enough to start)	Writes the Amharic summary and plain-language warnings.	See 7.7: it writes words, it never does arithmetic.

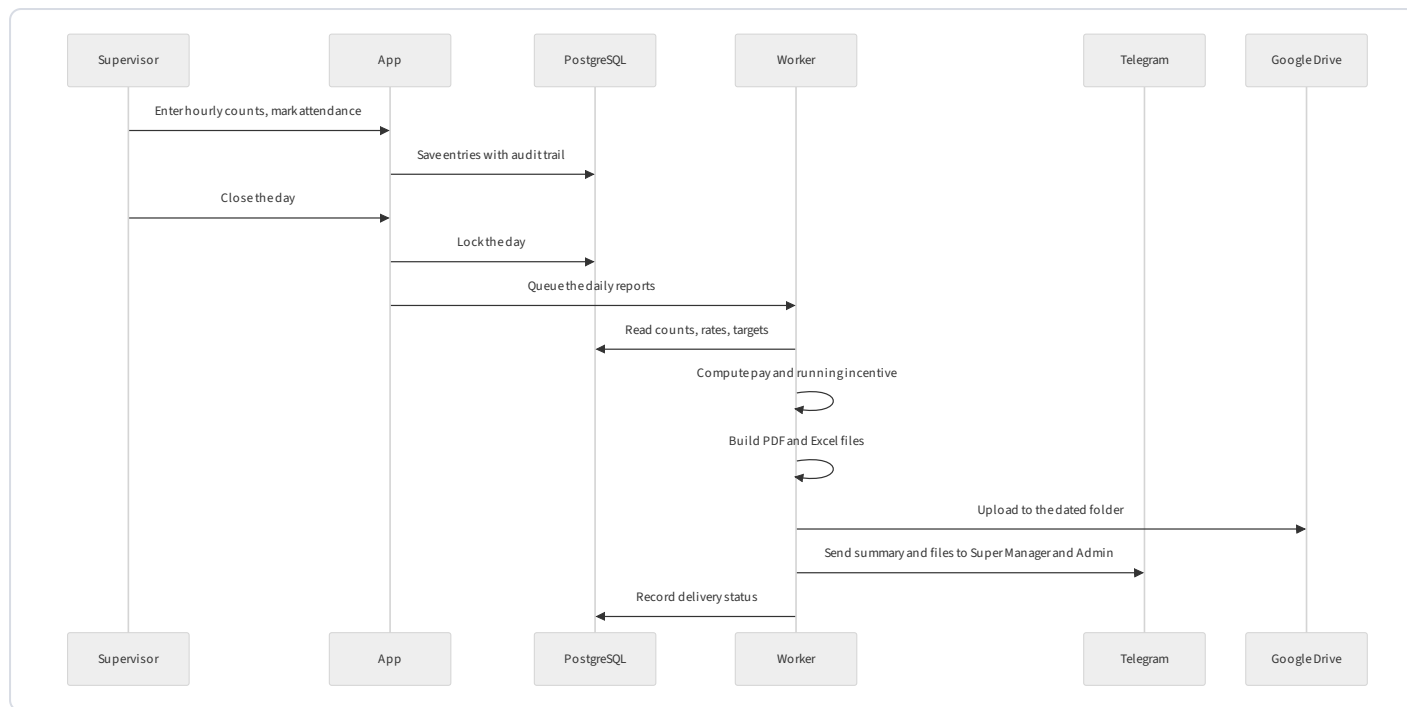
7.3 Hosting options

Prices change often, so this compares technical fit only. Cost comparison belongs in the client-facing cost document.

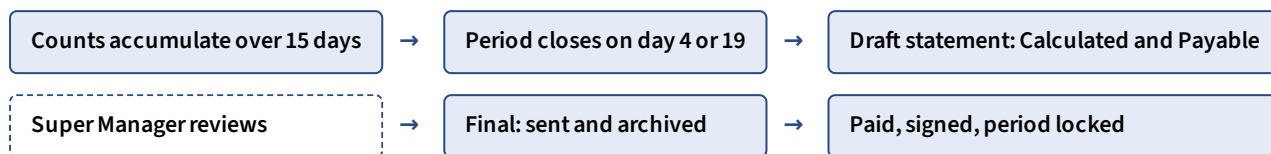
Option	Good for	Watch out for
A. Managed platform (for example Vercel) with a managed PostgreSQL	Fastest start, low ops effort.	Function time limits and cron limits, running a headless browser for PDFs, terms that may forbid commercial use on free plans, data outside Ethiopia, latency.
B. One virtual server abroad running containers (app, database, worker)	Cron, PDF generation and queue in one place. Predictable behavior.	You handle updates, backups and security. Latency to Addis Ababa.
C. Ethiopian cloud or hosting provider	Local payment, data in the country, lower latency.	Verify Node.js, containers, PostgreSQL, backups, and stable outbound access to Telegram and Google before committing.
D. Small server inside the factory	Keeps working on the local network during internet outages.	Needs power backup, physical security, remote access and someone to maintain it.

Recommendation. Build everything as containers so the host can change later. Start on B or C. Before choosing, test from the candidate host that api.telegram.org and Google APIs are reachable reliably, because the automation depends on them.

7.4 One day through the system



7.5 Incentive period lifecycle



If the reviewer finds a problem, the statement goes back for a correction with a reason and returns for review.

7.6 Amharic and Ethiopian-calendar requirements

- **Strings:** all screen text lives in a translation file, never hard-coded, and follows the glossary in chapter 14 so the same term is used everywhere.
- **Fonts:** an Ethiopic font (such as Noto Sans Ethiopic) in the UI and embedded in every PDF. Test full character coverage on real names.
- **Calendar:** an Ethiopian-calendar date picker; conversion at the edges only; unit tests around Pagume, leap years and period boundaries.
- **Numbers:** Arabic digits, as in the samples (confirm in [Q15](#)).
- **Sorting and search:** Amharic-aware ordering and search by any part of a name.
- **Telegram tables:** Ethiopic characters do not align in monospace text, so tables are sent as PDF or Excel files and the message stays a short summary.

7.7 AI for the daily summary

- The system computes every number first. The model receives a structured set of facts and only writes the Amharic narrative.
- The output is checked: every number in the text must match the facts. If the model fails or a number does not match, a fixed Amharic template is sent instead.
- Send aggregates, not employee names or wages, to the outside model unless the client agrees otherwise.

- Keep prompts and outputs logged. A person reviews the wording during the first weeks. The model provider stays replaceable.

7.8 Reliability

- **Connection and power:** entries are saved as drafts on the device immediately and retried automatically. Full offline queuing arrives in phase 3. Plan a charging station and backup power for the router.
- **Delivery outbox:** every report is a row with status (pending, sent, failed) and retries with backoff. If Telegram or Drive is down, nothing is lost and the Super Manager or Admin can download from the app.
- **Backups:** nightly encrypted database dump to a second location, plus a monthly restore test. A backup never restored is not a backup.
- **Monitoring:** a health check and a message to a developer chat when jobs fail.

7.9 Security and privacy

- Salary and incentive data is sensitive, and fixed salaries most of all. They are visible only to the Super Manager and the Admin, with field-level checks on the server. Salary and incentive files go only to their Telegram chats, which are bound to their accounts with a one-time code. The bot ignores every other chat.
- Telegram bot chats are not end-to-end encrypted. If that is a concern, send a summary with a Drive link and keep full statements and salary files in a private Drive folder only.
- Admin and Super Manager actions on other departments' data are marked in the audit log. Admin changes to salaries, targets or incentive rates notify the Super Manager.
- Role checks on the server, least-privilege access to Drive, secrets in environment variables, HTTPS only, database not exposed to the internet, encrypted backups.
- Collect only the employee data the system needs, and follow local data-protection rules for notice and consent.

8. Automation and reports

Every report is generated by the worker, sent to Telegram, saved to Google Drive and recorded in the database, all from one job.

Report	Trigger	Content	Format	Telegram to
Daily Amharic summary	Day close, with a fixed-time fallback	Headline numbers, alerts, orders at risk	Message and PDF	Super Manager, Admin, department managers
Daily production sheet	Day close	Layout of salary_schedule.pdf without pay: target, produced, difference and percent of target per worker	PDF and Excel	Super Manager, Admin, Production Manager
Daily piece-count sheet	Day close	Eight hourly columns and daily +/- per worker	PDF	Super Manager, Admin, Production Manager
Running incentive	Day close	Period to date: +, -, mistakes, calculated and payable per worker	PDF and Excel	Super Manager, Admin only
Incentive statement	Period close, day 4 and day 19, after Super Manager approval	Layout of the incentive_v4.pdf statement pages, with Calculated and Payable	PDF and Excel	Super Manager, Admin only
Monthly incentive summary	After the day-19 statement	Department table and bar chart, using Payable	PDF	Super Manager, Admin only
Monthly fixed salary schedule	Month end, after Super Manager approval (date to be agreed)	Name, department, fixed salary, days attended, signature column	PDF and Excel	Super Manager, Admin only

Report	Trigger	Content	Format	Telegram to
Daily inventory report	Morning and day close	Stock by SKU, movements, items below minimum	PDF or Excel	Super Manager, Admin, Store Keeper
Cutting wastage report	Day close	Each cut job, wastage % against the limit	PDF	Super Manager, Admin, Cutting, Production
Employee productivity	Day close	Hours against output, efficiency against target	PDF	Super Manager, Admin, Production, HR
Order status	Day close and on request	Each order by stage, delays	PDF	Super Manager, Admin, Production
Instant alerts	On event	Stock, wastage, delayed order	Short text	By alert rule

Drive folder structure and file names

Folders follow the Ethiopian calendar because that is how the client reads dates: **Factory Reports / year / month / report type**. Salary and incentive folders are shared only with the Super Manager and the Admin. File names start with the date and are zero-padded so they sort correctly, for example **2018-12-15_daily-production-sheet_v1.pdf**. A regenerated report gets **_v2**, and the earlier file is kept.

Telegram setup

- Three destinations: a management chat for the Super Manager and the Admin (summary, alerts, salary and incentive files), a department managers' group (summary and alerts, no salary or incentive data), and a developer chat (system errors).
- Each person links their Telegram account with a one-time code shown in the app. Recipients and the reports they may receive are managed by the Admin and visible to the Super Manager.

Draft Amharic message templates

Wording for the client to review. Placeholders in braces are filled by the system.



የብክነት ማስጠንቀቂያ

ትዕዛዝ: {order} ስታይል: {style}

የወጣ ጨርቅ: {kg} ኪ.ግ | የተቆረጠ: {pcs} ፍሬ

ብክነት: {pct}% (የተፈቀደ: 5%)

የቆረጣኝ ስም: {name}



የክምችት ማስጠንቀቂያ

{material} ({sku}) ክምችት ከዝቅተኛ ወሰን በታች ነው።

ያለ: {qty} {unit} | ዝቅተኛ ወሰን: {min} {unit}



የዕለት ማጠቃለያ - {date}



የተቆረጠ: {n} ፍሬ | ብክነት: {pct}%



የተሰፋ: {n} ፍሬ



ጥራት ያለፈ: {n} | የተመለሰ: {n}



የታሸገ: {n} | የተላከ: {n}



ከዲላማ በላይ የሰሩ ሰራተኞች: {n} ከ {total}



ማስጠንቀቂያዎች: {n}

The daily summary no longer shows a daily pay total, because salary is fixed. It shows how many workers finished above target instead. Incentive amounts are sent only in the Super Manager and Admin files.

9. Screens by role

Role	Main screens	Design notes
Super Manager	Dashboard with graphs and profit and loss indicators, approval queue (incentive statements, salary schedules, salary and rate changes), every department screen, reports and delivery status	Approvals first. One tap to see what changed since the last period.
Admin	Users and permissions, settings and alert rules, Telegram and Drive connections, backups, audit log, corrections after lock	Every correction asks for a reason. Pay-related corrections show a notice that the Super Manager will be told.
Store Keeper	Receive materials (scan or enter, supplier, lot, quantity), issue to cutting against a request, stock levels with low-stock flags	SKU assigned by the system. Scan first, type last.
Cutting Manager	Fabric request, cut record (order, lot, weight used, pieces cut, bundles), wastage result	Wastage % appears instantly and turns red above the limit.
Production Manager	Live line board (bundles by stage), hourly count grid by department, day close, + and – adjustments, verification of operator entries, delayed orders	Grid of workers by 8 hours with a number pad, target prefilled, one tap to copy the previous hour.
QC Inspector	Scan bundle or garment, pass or fail, defect reason, responsible operation, send back to line	Three taps for a pass. Defect reasons come from a fixed list.
Finished Goods Manager	Staging queue, tag or barcode print, dispatch note	Order status updates automatically on dispatch.
HR / Clerk	Employee list (no salary), attendance and hours, line assignment	Bulk attendance for a whole line at once.
Shop Floor Operator	My work today, enter my count, bundle in hand	Very few fields, large buttons, drafts survive lost signal.

Rules for every screen: Amharic only; numeric keypads for numbers; defaults instead of typing; confirmation before irreversible actions; drafts saved automatically; clear error messages that say what to do next.

10. Roadmap and testing

Three phases, ordered by dependency and by where the money is. Phase 1 replaces the paper forms that decide the incentive, because those outputs are already specified by the sample PDFs. If the client's requirements document defines its own phase boundaries, align these phases to them; the content stays the same.

Phase	Content	Done when	Rough effort, one developer
0. Confirm	Settle the open questions in chapter 13, collect a sample fixed salary schedule, the real incentive card and the employee list, agree the Amharic glossary.	Written answers to Q1 to Q15.	About 1 week
1. Incentive and floor core	Login and the nine roles (including Super Manager and Admin), Amharic UI and fonts, Ethiopian calendar, employee and department master with fixed salaries, incentive card, attendance, hourly counts, day close, daily production sheet, incentive periods and statements (Calculated and Payable), penalty ladder, approval flow, monthly fixed salary schedule, monthly summary and chart, Telegram delivery, Drive archive, audit log, backups	Sample incentive statements reproduced exactly from entered data. One line piloted for a full month.	5 to 7 weeks
2. Materials to packing	Store keeper and stock, SKU and lots, BOM, cutting and wastage, bundles through all stages, QC and rework loop, alerts, four core reports, order status	A full order can be tracked from receiving to packing.	6 to 8 weeks

Phase	Content	Done when	Rough effort, one developer
3. Insight and hardening	Finished goods and delivery, barcode and tags, dashboards and graphs, AI Amharic summary, delayed-order alerts, offline queue, load and user acceptance tests, training	Parallel run passes. Client signs off.	4 to 6 weeks

Effort figures are rough and depend on how quickly the client answers the open questions.

Build order inside phase 1

1. Application skeleton, login, the nine roles and permission checks, Amharic strings, fonts and calendar conversion.
2. Departments, incentive card, employees and fixed salaries, loaded from the sample data.
3. Attendance and hourly count entry with day close.
4. Incentive engine as a standalone module of pure calculations, tested against the sample statements before any screen uses it.
5. PDF and Excel layouts that match the samples, plus the Payable column.
6. Approval flow, then Telegram and Drive delivery with the outbox and retry.
7. Audit log, locking, corrections, backups.
8. Pilot on one line with real supervisors.

Acceptance tests taken from the samples

- Daily production sheet: each of the 53 workers with a target shows produced against target (for example 525 against 520 gives +5), the day totals 22,679 pieces and 40 of 53 workers are at or above target.
- Incentive statements: 67 rows per period; Calculated totals are 3,880.50 ETB (+4,865, -430) and 3,765.00 ETB (+4,610, -270), month 7,645.50 ETB; Payable totals are 4,243.50, 4,009.00 and 8,252.50 ETB.
- Monthly summary: the 24 department rows in the appendix, Calculated and Payable, including the negative department totals.
- Fixed salary schedule: layout agreed against a client sample once one exists.
- Roles: the Super Manager and the Admin can open every screen; only the Super Manager can approve a statement; department roles cannot see salaries; an Admin correction on pay data sends a notice to the Super Manager.
- Wastage: the 5.1 example gives 6.0% and triggers an alert; 5.0% exactly does not.
- Amharic: every PDF shows all characters correctly; a period spanning Pagume produces sensible boundaries.

Rollout

- **Parallel run:** keep the paper forms for at least one full month (both pay periods) and compare totals per worker. Go live only when every difference is explained.
- **Training:** short sessions per role in Amharic, with a one-page guide per screen. Start with supervisors, since they enter the counts that decide the incentive.
- **Support:** a named contact and a fast way to report a wrong number during the first month.

11. Risks

Risk	Why it matters	Response
Incentive rules unclear or changing	A wrong incentive destroys trust in the system.	Close the open questions first, keep rates and rules configurable, run in parallel with paper.
Only one Admin and one Super Manager	A lost phone, illness or resignation could lock everyone out, or leave one person with unchecked power.	Recovery codes held by the owner, a named deputy, two-key approval on money, audit log (recommendations R6 and R8).
Supervisors keep using paper	Data is incomplete.	Make entry faster than paper, print the familiar layouts, pilot on one line, train supervisors first.
Data-entry load (about 540 hourly cells per day)	Delays and typing errors.	Department grid, number pad, prefilled targets. If the client agrees, capture two or three checkpoints per day instead of hourly.
Incentive cost creeping up	Most sample workers finish above target, so cost can grow quietly.	Dashboard of share above target and incentive cost per 1,000 pieces; targets reviewed each quarter (R2).
Internet or power interruptions	Lost or late entries.	Drafts on the device, automatic retry, offline queue in phase 3, backup power for the router and tablets.
Amharic quality in PDFs and AI text	Broken glyphs or wrong wording.	Embedded fonts, PDF snapshot tests, template fallback, human review of AI text.
Scope creep toward a full ERP	Nothing gets finished.	Phases, out-of-scope list, written change requests.
Single developer	Client depends on one person.	Simple stack, containers, this documentation, a written runbook, tested backups, client-owned accounts.
Third-party limits or changes (Telegram, Google, AI free tier)	Reports could stop.	Outbox and retries, failure alerts, replaceable providers, download from the app as fallback.
Salary and incentive data exposure	Sensitive personal and financial information.	Field-level access, restricted chats and folders, encrypted backups, audit log (see 7.9).

12. Recommendations

These are my recommendations across the whole project, from pay design to contracts. They are advice, not requirements: the client and you decide. Items are grouped by theme and marked with the moment they matter.

If you do only five things: get written answers to Q1 to Q8 (R21); build and test the incentive engine against the sample statements before any screen (R10); use two keys on money (R6); protect fixed salaries above everything else (R7); run one full month in parallel with paper (R17).

Pay and incentive design

#	Recommendation	Why it matters	When
R1	Pay incentive only on quality-passed pieces	With a fixed salary the incentive is the main lever. If rejected pieces count, speed will beat quality. Phase 1 relies on supervisors excluding known rework; Phase 2 ties counts to QC records per bundle.	Phase 1, then 2

#	Recommendation	Why it matters	When
R2	Review targets every quarter, with data	In the sample statements, 45 of the 61 workers who have a target finished period 1 above it, and 43 finished period 2 above it. That may mean the program works, or that targets are comfortable. Track the share above target and the incentive cost per 1,000 pieces, and change targets only at the start of a period.	Phase 1 dashboard
R3	Show the cost of the "payable never below zero" rule before approving it	Flooring negatives at zero raises the sample month from 7,645.50 to 8,252.50 ETB, about 8% more. The Super Manager should see this number before choosing the rule.	Before build (Q2)
R4	Consider a team bonus for support roles	Helpers, QC and line control have target 0, so they always earn 0 in the sample, even in a very good month. A line-level bonus tied to line output or QC pass rate is an option. It is the client's call.	Later
R5	Let workers see and dispute their count	Show each worker their running incentive (client choice, Q15) and allow a short dispute window, for example 48 hours after the day closes and before the period locks. Most pay arguments are count arguments, and transparency prevents them.	Phase 1

Control and security

#	Recommendation	Why it matters	When
R6	Use two keys on money	Supervisor enters, Production Manager verifies, Super Manager approves. The Admin may correct with a reason, and every Admin correction on pay data notifies the Super Manager. No one approves their own entries.	Phase 1
R7	Protect fixed salaries more than anything else	Field-level access (Super Manager and Admin only), never in group chats, private Drive folders, encrypted backups. Consider sending only a Drive link on Telegram for salary files.	Phase 1
R8	Plan for the one-Admin, one-Super-Manager setup failing	With one person per top role, a lost phone or a resignation locks everyone out. Keep sealed recovery codes with the owner, write a reset procedure, and name a deputy who can be promoted in an emergency.	Before go-live
R9	Check counts against a second source	Compare each department's worker counts with bundle stage logs and QC results, flag large gaps, and audit a few workers each month. This is also how a "mistake" can be proven fairly.	Phase 2

Build and technology

#	Recommendation	Why it matters	When
R10	Build and test the incentive engine first	Implement the calculation as isolated, pure functions and test it against the sample statements (appendix values) before any screen exists. If the numbers are right, everything built on them is safe.	Phase 1 start
R11	Keep the first release small	Phase 1 is employees, attendance, counts, incentive, fixed salary schedule, Telegram and Drive. Pilot on one line for a full month before adding stock and cutting.	Phase 1
R12	Start with supervisor entry, add scanning later	Barcode scanning is more accurate but adds steps for workers. Introduce bundle scanning once the counts flow is trusted.	Phase 3
R13	Choose hosting after a reachability test	From each candidate host, confirm that Telegram and Google APIs are reachable reliably. Run everything in containers. Do not run the client's production system on a free plan whose terms forbid commercial use.	Before build
R14	Design for weak internet and power	Drafts saved on the device, automatic retry, an offline queue in Phase 3, a small UPS for the router and a charging station for the tablets.	Phase 1 and 3
R15	Keep backups you have actually restored	Nightly encrypted dump to a second location, a monthly restore test on a fresh machine, and a written runbook.	Phase 1
R16	Have the client's Amharic speakers review the wording	Review the glossary, screen labels and Telegram templates with two or three staff. Keep the fixed-template fallback for the AI summary, and send only aggregate data to the AI service.	Phase 1 and 3

Rollout, people and contract

#	Recommendation	Why it matters	When
R17	Run in parallel with paper for one full month	Compare per-worker totals for both pay periods. Go live only when every difference is explained.	Phase 3 (pilot earlier)
R18	Train supervisors first and name a champion on the client side	Supervisors enter the counts that decide the incentive. One trusted person at the client answers daily questions and collects change requests.	Phase 1
R19	Check the physical setup early	Wi-Fi coverage on the floor, 8 to 10 inch tablets with rugged cases, chargers, and a safe place to leave them at night.	Before pilot
R20	Put the client's name on every account	Hosting, domain, Google Drive, Telegram bot and code repository should belong to the client, with you as an authorised user. It avoids disputes at handover and lets the client continue if your availability changes.	Before build
R21	Agree scope and support in writing	Use this document as the baseline. List phases with acceptance tests, a change-request process, a support period and who pays running costs. Keep the client's written answers to the open questions with it.	Before build
R22	Ask HR or a labour-law adviser to review the incentive and penalty rules	Incentives, suspensions and warnings can be regulated. I am not a lawyer, so this should be checked by someone qualified before the pilot.	Before pilot

13. Open questions for the client

Each question includes the default I would build until the client answers. None blocks the start of the project, but Q1 to Q8 must be answered before the incentive engine goes into a pilot. Q1, Q2 and Q11 are new in version 2. Q3 replaces two earlier questions about rates and targets. The others carry over from version 1 with new numbers.

#	Question	Why it matters	Default until answered
Q1	How is the fixed salary handled? Does the system only store and print it, or also handle absence deductions, overtime, tax and pension? Who sets salaries?	Version 1 left base payroll out of scope. Any calculation beyond printing a fixed amount adds rules and legal duties. There is also no sample of a fixed salary schedule yet.	Store the fixed salary per employee, print the monthly schedule with days attended for reference, calculate nothing else. The Super Manager sets salaries.
Q2	Shortfalls: is the period's surplus netted against its shortfalls (as the sheet's formula does) or does only surplus count? Is the payable amount floored at zero, and is anything ever deducted?	The sample shows negatives (−50.00, −27.00). Flooring at zero raises the sample month from 7,645.50 to 8,252.50 ETB. Wage deductions may be regulated by labour law.	Net per period as in the sheet, payable never below 0, nothing deducted from the fixed salary. Statements show Calculated and Payable.
Q3	Is the incentive card in incentive_v4.pdf the single source of targets and rates, and how do the 13 operations on the daily production sheet map to its 21 rows?	The two PDFs disagree on targets for the same work (hit press 82 vs 90 per hour, ironing 41 vs 45).	Yes: one incentive card with dated targets and rates, plus a mapping table between operations and departments.
Q4	Are +pieces and −pieces the amount above and below target, and do the eight hourly cells hold actual pieces or differences?	Decides what supervisors type and how errors are caught. In the sample no worker has both + and − in a period, so the columns may already be period results.	Enter actual pieces per hour; the system derives differences and totals.
Q5	If a worker is absent or works part of a day, is the target reduced in proportion?	Otherwise absence produces a large negative.	Target = hourly target × hours attended. No penalty for approved absence.
Q6	What exactly is a "mistake" (ማሳሰብ), who records it and how is it verified?	It counts double and starts the penalty ladder. The sample column is always 0.	Entered by supervisor or admin with a reason, checked against bundle and QC records, logged.

#	Question	Why it matters	Default until answered
Q7	How are offences counted over time, and what is "2 months" in pay periods?	Decides when a suspension starts and ends.	Admin-controlled rolling count; 2 months = 4 periods; 3rd and 4th offences only open HR cases.
Q8	How are periods handled around Pagume and public holidays?	Day 20 of "the previous month" does not exist when that month is Pagume.	Period boundaries configured per month.
Q9	How is cutting wastage measured, and is the 5% limit strict and per what?	The alert and the report depend on it.	Per cut job, from the standard weight per piece, alert above 5%, limit editable per style.
Q10	Is the stage order fixed as listed? Is there a final inspection, and are stages skipped for some styles?	Bundle routing and progress views depend on it.	Route per style as an ordered list, default the nine listed stages.
Q11	Super Manager and Admin: who holds each role (is the owner one of them)? May the Admin approve pay, or only the Super Manager? May either change counts or salaries, and who is told when they do?	Both are meant to control everything, which is powerful and needs a clear rule about who signs off money.	Both see and can act on everything. Only the Super Manager approves incentive statements, salary schedules and salary changes. Admin corrections on pay data notify the Super Manager and appear in the audit log.
Q12	Who enters counts for trimming, hit press, ironing and packing (no matching role among the nine)? Is the "supervisor" on the forms the Production Manager? Do operators enter their own counts?	The incentive depends on the count, so the source of the count matters.	Supervisors enter or verify; operator entries optional and pending until verified.
Q13	Do incentive rates also differ by garment type on combined operations (for example side seam T-shirt 0.90 vs trouser 0.80)?	The old salary sheet note gave different piece rates for T-shirt and trouser on operations 8, 9 and 10. The incentive card has one rate per department.	One rate per department. If the client wants garment-based rates, count lines carry the garment type.
Q14	How do QC rejections and repairs change counts, and who is credited for a repair?	Rework is excluded from incentive.	A rejected piece is removed from the responsible operation's count; repaired pieces are credited to the repair department only.
Q15	Output conventions: keep the signature column, Ethiopian dates and Arabic digits? May workers see their own running incentive? Who may see fixed salaries (does HR need to)? Who receives salary and incentive files?	Layout fidelity and privacy.	Signature column kept blank, Ethiopian dates, Arabic digits, worker self-view off, salaries visible to the Super Manager and Admin only, salary and incentive files to their chats only.

14. Glossary

Amharic or term	Meaning
ኢንሴንቲቭ	Incentive: extra pay, on top of the fixed salary, for output above target
ቋሚ ደሞዝ	Fixed salary
ሱፐር ማኔጀር, ዋና ሥራ አስኪያጅ	Super Manager: the top business role that manages all other roles
አስተዳዳሪ	Admin: controls the system and its users
ደሞዝ ሰሌዳ	Salary schedule (payroll sheet)
የቁራጭ ክፍያ	Piece rate: pay per piece (used on the old salary sheet; no longer used for pay)
ሳንቲም	Literally cent. On the incentive sheet: the rate in ETB per piece
ፍሬ	Piece (counting unit for garments and operations)
+ፍሬ, -ፍሬ	Pieces above target, pieces below target (my reading)
ቧላማ	Target
ማሳሳት	A mistake or misreport of the count (my reading, Q6)

Amharic or term	Meaning
ቆጠራ	Count, tally
ቆረጣ	Cutting
ስፊት	Sewing
ለቀማ, ቅንጫባ	Trimming: picking off loose threads
ካውያ	Ironing
ማሸግ, ፓኪንግ	Packing
ብክነት	Waste, wastage
ክምችት	Stock, inventory
መጋዘን	Warehouse, store
ባንድል	Bundle: a group of cut pieces that travels together
ሶፐርቫይዘር	Supervisor
ፊርማ	Signature
ጠቅላላ	Total
ተ.ቁ	Serial number
Pagume (ጳጉሜ)	The 13th month, 5 or 6 days long
E.C.	Ethiopian Calendar
BOM	Bill of materials: how much of each material one garment needs
SKU	Stock keeping unit: the unique code of a material or product
Hitpress, DTF	Heat press machine; DTF is a printed-transfer sticker applied with it
Overlock, Interlock, Single needle, Kanshay	Sewing machine types named in the samples

Appendix: sample reference values

These come from incentive_v4.pdf and serve as test data for the incentive engine. English department names are approximate. The Payable column applies the version 2 rule: each worker's period result is floored at zero before adding up.

A. Incentive card (21 rows), the only rate table

#	Department	Machine	Target per hour	Rate (ETB per piece)	Workers
1	Shoulder, back and front (ትከሻ ፍሬንት)	Overlock	70	1.00	4
2	Waist tacking (ወገብ መቀምቀም)	Overlock	70	1.00	1
3	Sleeve and side (እጅ / ሳይድ)	Overlock	75	1.00	4
4	Pocket attaching (ኪስ መለጠፍ)	Single	80	0.90	4
5	Side (ሳይድ)	Overlock	78	1.00	4
6	Sleeve (እጅት)	Overlock	76	1.00	4
7	Sleeve topstitch and badge (እጅት ደርዝ እና ባጅ)	Single	82	0.90	2
8	Interlock hem (Interlock ማጠፍ)	Interlock	72	1.20	4
9	Waistband joining (ወገብ መቀጠም)	Single	75	0.90	4
10	Kanshay waist prep (የካንሻይ ወገብ ዝግጅት)	Single	82	0.80	3
11	Heat-press applying (ሂትፕረስ መለጠፍ)	Hitpress	90	0.80	4

#	Department	Machine	Target per hour	Rate (ETB per piece)	Workers
12	Ironing (ካውያ)	Kawya	45	0.80	4
13	Thread trimming (ቅንጫባ)	none	30	0.70	8
14	Packing (ማሸግ)	none	55	0.60	5
15	Sticker and elastic cutting (ስቲከር / ላስቲክ መቀጥ)	none	60	0.80	1
16	Damage and repair (ዳሜጅ / ስራ መስረት)	none	60	0.80	2
17	Helper, trouser cutting (ረዳት / ስራ መቁረጥ)	none	50	0.80	1
18	Cutter (ቆራጭ)	none	65	0.90	1
19	Assistant cutter (ረዳት ቆራጭ)	none	55	0.90	1
20	Quality inspector	none	0	0.60	1
21	Helper (ረዳት)	none	0	0.60	1

The remaining sample workers (line helpers, line running and control) have no target and no rate and always earn 0.

B. Monthly incentive summary by department

#	Department	Workers	Day 4, calculated (ETB)	Day 19, calculated (ETB)	Month, calculated (ETB)	Month, payable (ETB)
1	Shoulder, back and front	4	560.00	135.00	695.00	755.00
2	Waist tacking	1	150.00	180.00	330.00	330.00
3	Sleeve and side	4	350.00	350.00	700.00	700.00
4	Pocket attaching	4	85.50	423.00	508.50	535.50
5	Side	4	270.00	380.00	650.00	700.00
6	Sleeve	4	320.00	-70.00	250.00	400.00
7	Sleeve topstitch and badge	2	189.00	40.50	229.50	229.50
8	Interlock hem	4	264.00	408.00	672.00	672.00
9	Waistband joining	4	144.00	189.00	333.00	414.00
10	Kanshay waist prep	3	344.00	120.00	464.00	464.00
11	Line helpers	2	0.00	0.00	0.00	0.00
12	Heat-press applying	4	156.00	200.00	356.00	396.00
13	Ironing	4	392.00	408.00	800.00	800.00
14	Thread trimming	8	329.00	542.50	871.50	920.50
15	Packing	5	297.00	177.00	474.00	504.00
16	Sticker and elastic cutting	1	-24.00	72.00	48.00	72.00
17	Damage and repair	2	-32.00	-24.00	-56.00	0.00
18	Helper, trouser cutting	1	-40.00	0.00	-40.00	0.00
19	Cutter	1	81.00	180.00	261.00	261.00
20	Assistant cutter	1	45.00	54.00	99.00	99.00
21	Quality inspector	1	0.00	0.00	0.00	0.00
22	Helper	1	0.00	0.00	0.00	0.00
23	Line running	1	0.00	0.00	0.00	0.00
24	Line control	1	0.00	0.00	0.00	0.00
Total		67	3,880.50	3,765.00	7,645.50	8,252.50